



Jersey City Bitcoin Socratic Seminar #7

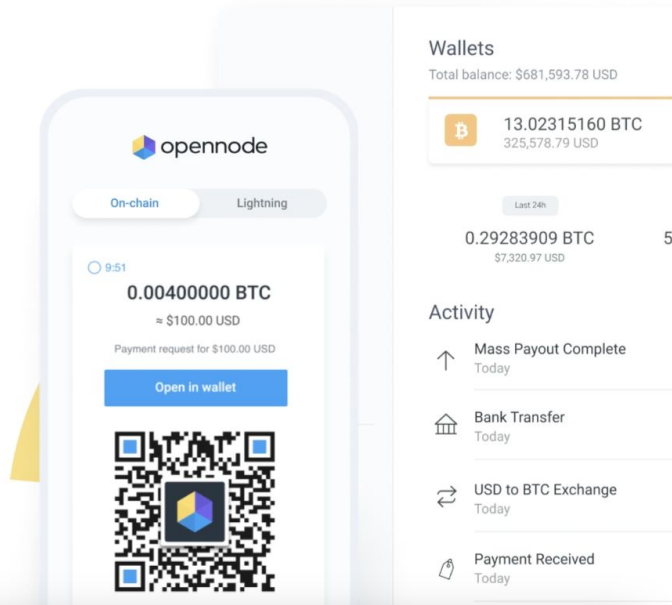
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Agenda:

- Intro
- Grassroots Recap
- Marketing
 - All about downtown!
 - Guerrilla marketing initiative
- BTC Vegas Roundtable
- Macro Update
- Bitcoin Price Update
- Bitcoin History
- Bitcoin News
- How does bitcoin work? Mining Prerequisites
- Bitcoiner Values
- Book Sharing Program



JC Bitcoin Meetup General Guidelines

- Basic Rules:

- Please no photos, no videos, no public tagging without consent!
- This group is Bitcoin only so please limit discussion of other cryptocurrencies.
- PLEASE respect the space. We love the barrow mansion and ask for help making it look better than when we arrived.
- For newcomers and OGs alike it is worth reiterating that absolutely no one from JC Bitcoin will ever solicit Bitcoin or crypto transfers here or on any other channel

Number Time

Clark Moody Bitcoin		Dashboard	
Markets ?		Blockchain ?	
Price	\$74,700	Block Height	945,130
Sats per Dollar	1,339	UTXO Set Size	165,209,175
Market Capitalization	\$1.50T	Block Time, Prior Year	9:59
Check out Dashboard v2		Chain Size	836.0 GB
All-Time High ?		OP_RETURN Data	0.0 GB
All-Time High Price	\$126,160	Bitcoin Network ?	
Decline from ATH	-40.80%	Reachable Bitcoin Nodes	20,969
ATH Date	October 6, 2025	Bitcoin Tor Nodes	12,385
Days Since ATH	190	Percentage Tor Nodes	59.06%
Gold ?		Lightning Network (Public) ?	
Bitcoin priced in Gold	17.7 oz	Total Capacity	4,764.34 BTC
Bitcoin vs Gold Market Cap	4.98%	Capacity Value	\$355.9M
Corporate Treasuries ?		Total Nodes	9,997
Held in Corp. Treasuries	2,285,604 BTC	Total Channels	39,351
Value in Corp. Treasuries	\$170.7B	Tor Capacity	3,664.30 BTC
Supply Pct. in Corp. Treasuries	11.42%	Percentage Tor Capacity	76.9%
Supply ?		Tor Nodes	5,406
Money Supply	20,015,804.28 BTC	Bitfinex Lightning Node ?	
Percentage Issued	95.31%	Inbound Capacity	252.20 BTC
Unspendable	230.10 BTC	Outbound Capacity	818.89 BTC
Issuance Remaining	983,965.60 BTC	Bitfinex Channels	561
		Liquid Sidechain ?	
		Peg-in Capacity	-178.28 BTC
		Peg-in Capacity Value	\$-13,317,759.77
		Liquid Block Height	3,790,025
		Liquid Chain Size	56.2 GB
		Transactions ?	
		Total All Time	1,339,713,701
		Rate, 30 days	6.3 tx/s
		Count, 30 days	16,312,784
		Chain Security ?	
		Hash Rate, 90 Days	983.7 EH/s
		Chain Work	96.1 bits
		Chain Rewrite Days	1,007
		Annual Mining Revenue	\$16.1B
		Mining ?	
		Hash Rate, 2016 Blocks	976.0 EH/s
		Difficulty	139.0x10 ¹²
		Difficulty Epoch	469
		Last Difficulty Change	3.9%
		Block Time, 2016 Blocks	10:07
		Difficulty Retarget ?	
		Blocks to Retarget	374
		Retarget Date	April 17, 2026
		Estimated Difficulty Change	-2.4%
		Block Time, Diff. Epoch	10:14

Quick Mempool Check

mempool.space

Explore the full Bitcoin ecosystem

0 sat/vB	-0 sat/vB	-0 sat/vB	-0 sat/vB	-0 sat/vB	-0 sat/vB	-1 sat/vB
0.12 - 0.13 sat/vB	0.13 - 0.13 sat/vB	0.13 - 0.14 sat/vB	0.13 - 0.14 sat/vB	0.14 - 0.14 sat/vB	0.14 - 0.15 sat/vB	0.15 - 251 sat/vB
0.001 BTC	0.001 BTC	0.001 BTC	0.001 BTC	0.001 BTC	0.001 BTC	0.029 BTC
4,684 transactions	7,967 transactions	7,698 transactions	6,516 transactions	5,798 transactions	4,321 transactions	
In ~61 minutes	In ~51 minutes	In ~40 minutes	In ~40 minutes	In ~30 minutes	In ~20 minutes	In ~10 minutes

945130	945129	945128	945127	945126	945125	945124
-0 sat/vB	-2 sat/vB	-1 sat/vB	-0 sat/vB	-1 sat/vB	-0 sat/vB	-0 sat/vB
0.14 - 316 sat/vB	1.00 - 100 sat/vB	0.15 - 100 sat/vB	0.14 - 78.0 sat/vB	0.14 - 100 sat/vB	0.14 - 26.2 sat/vB	0.14 - 26.2 sat/vB
0.008 BTC	0.025 BTC	0.012 BTC	0.008 BTC	0.016 BTC	0.003 BTC	0.003 BTC
5,667 transactions	3,594 transactions	4,022 transactions	4,434 transactions	4,066 transactions	3,028 transactions	3,028 transactions
15 minutes ago	13 minutes ago	47 minutes ago	58 minutes ago	63 minutes ago	79 minutes ago	81 minutes ago
AntPool	SpiderPool	Foundry USA	Foundry USA	SECPool	Foundry USA	Foundry USA

TRANSACTION FEES			
No Priority	Low Priority	Medium Priority	High Priority
0.2 sat/vB	0.4 sat/vB	0.9 sat/vB	2 sat/vB
\$0.02	\$0.04	\$0.09	\$0.17

DIFFICULTY ADJUSTMENT

difficulty halving

~10.2 minutes

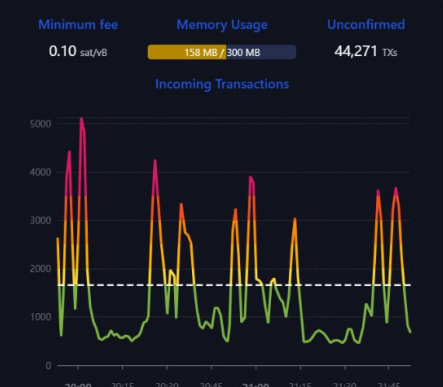
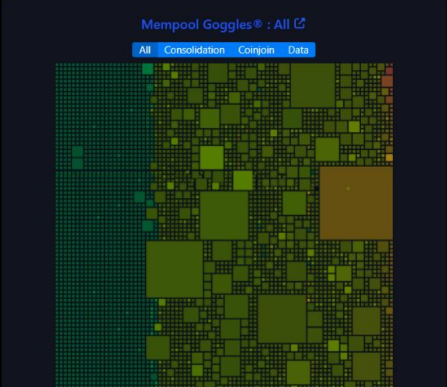
Average block time

2.35 %

Previous: 3.87 %

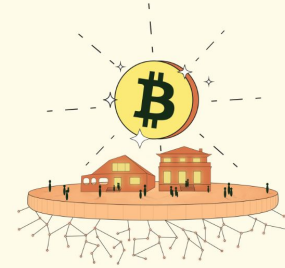
In ~2 days

April 17 at 1:45 PM



Bitcoin Park & Grassroots Summit

- Goal: bring back best practices, connections, and resources for JC Bitcoin
 - Bulletin board of services
- Who wasn't here last month?



GRASSROOTS
BITCOIN 2026

 bitcoin park

Bitcoin Park & Grassroots Summit

- Charlotte
- Portland
- Barcelona
- Philly
- Roatan Honduras
- Louisville
- Bitcoin Students Network
- Bitcoin Park
- Atlanta Bitplebs
- Albany
- Tolido Ohio
- Chicago
- LA Carl
- Lexington
- Peru
- Salt Lake
- BTC Bay
- Kentucky
- Kansas City
- Nebraska
- Chattanooga
- NYC



Bitcoin Park & Grassroots Summit

- Charlotte
 - 100 Businesses 1 BTC
- Portland
 - Bitcoin is for everyone Conference
- Barcelona
- Philly
 - 24/7 Telegram
- Roatan Honduras
 - Private City
- Bitcoin Students Network
- Atlanta Bitplebs
 - Hackerspace and radio hour
- Chicago
 - Restaurant and coworking space

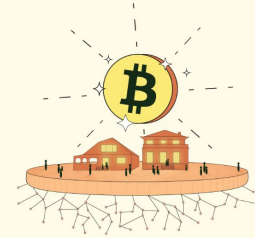


Bitcoin Park & Grassroots Summit

- LA Carl
 - Delete OPA
- Lexington
 - Focused on Merchant adoption
- Peru
 - Vendor markets, travel with bitcoin
- BTC Bay
 - Food kitchen, community gardens
 - Sound money sworey gala
 - Bitcoin education course University of tampa
 - Local politicians
 - Freedom tech meetups
 - Monthly support tiers
- Kansas City
 - Bitcoin Party,
 - Music art and culture curated event
 - Bitcoin block party, bitcoin summit
- Nebraska
 - Political action association like BPI
 - Family BBQ, county fairs
 - Building a bitcoin bank
- Chattanooga
 - Merchant adoption
 - Satsman gamifying bitcoin education
- NYC Freedom lab
- Salt Lake
 - Crypto/Bitcoin debate, bitcoin trivia

Bitcoin Park & Grassroots Summit

- Ideas:
 - Cake Wallet Monero Swaps
 - Guess the noce game
 - Non kyc local economy
 - Facebook Groups
 - 100 Biz 1 BTC
 - Zeppelin Hall
 - Butcher
 - Fat cash trades barrow basement
 - Signature events
 - Bitcoin & crypto debate
- What block height were we established?



GRASSROOTS
BITCOIN 2026

 bitcoin park

Marketing Initiatives

- Should we discuss marketing here?
- Strategy calls - Sunday Coffee shop
 - Sunday @ 11 (real coffee real people)
- Can someone print stickers/Fliers?
 - Bulletin boards, telephone poles



Marketing Ideas

- Openclaw?
- Challenge: every member brings ONE friend
- Helen's Partnership
- Fundraiser - Networking Event
- \$50 Yearly Membership - Satoshi Fellow
 - Resume builder
 - Merch!
 - Members Group Chat Access





hdsid_jc

🎵 half.cool · Dance Off (Instrumental)



22



hdsid_jc Less than 100 vendor spots left! 📅 The All About Downtown Street Fair is filling up fast — if your business wants to be part of Jersey City's biggest event of the year, now's your moment.



September 19th 2026, 12-8pm



Historic Downtown Jersey City



300 diverse vendors & thousands of attendees



Live Music , DJ sets, Kids rides, and so much more!

Spots go quick every year... don't miss your chance to be in the mix.



Apply now: jcdowntown.org (or link in bio)



M P 11:24



to bcc: me ▾

Hello!

Thank you for your application! We're excited to confirm that your spot for the 2026 All About Downtown Street Fair is secured.

We'll be sending out more details as the event approaches, so stay tuned. In the meantime, please be sure to submit your waiver for the event to complete your registration.

We're looking forward to seeing you at the Street Fair!

Maria



2026 All About
Downtown Stre...



PDF



2026 All About Downtown Street Fair

giancarnevale@gmail.com [Switch account](#)



Type of vendor space requested

Please select the option that best suits your business.

Vendor Categories

Non-Profit \$200

▼



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10x10

10x15

10x20

Color: **Canopy Tent**



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\$109.99



\$299.00

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Brand Dot4Dot

Material 600D Polyester Fabric, Commercial Hex Aluminum Frame

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\$8.99 delivery May 28 - 29.

[Details](#)

📍 Delivering to Long Branch 07740
- Update location

In stock

Usually ships within 4 to 5 days.

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[Customize now](#)

By clicking "Customize now", you agree to these [Terms and Conditions](#).

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Payment Secure transaction

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\$52.99



BITCOIN: FREEDOM. TECHNOLOGY. COMMUNITY.



Bitcoin



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What is Bitcoin?

A simple guide to the future of money.



Bitcoin

Secure.
Decentralized.
Permissionless.
For everyone.



Learn More



Scan the QR code for resources and getting started.



Bitcoin
Be your own bank.



Macro Update

- **Key Takeaway:**
 - Iran = the only macro variable right now
- Oil → inflation → yields → everything
 - Breakevens at 2.5%, multi-year highs
- 10Y at 4.5-4.75%
 - 30Y already >5%
- 5% on the 10Y is the breaking line
 - ~50% odds we get there (Bhatia)
- Gromen: yields get capped → Fed prints → 10-15% inflation in ~2 months
- Perry: don't short, wait for oil up + yields down combo
- No modeled Iran resolution



The collateral base is cracking at the same time

US Treasurys, Japan JGBs, and UK gilts are all repricing inflation, energy risk, and fiscal math.

MECHANISM

Yields rise -> collateral falls
Bank balance sheets take the hit

LONG SOVEREIGN STRESS

Hard prints where supplied. UK shown as stress signal.

US 30Y TREASURY

long-end collateral repricing

30Y HIGH
5.18%

JAPAN 30Y JGB

first time in history per brief

RECORD
>4%

UK GILTS

same direction; print not supplied

SIGNAL
stress

BITCOIN AT \$76,000 IS MISPRICED

Neutral money gets more valuable as collateral breaks.

SYSTEM STATE

6.9 / 12 *Watching*

DRIVEN BY

Treasury Market **L4.0**AI & Labor **L4.0**Sovereign **L2.5**

CONTAGION

Calibrating 4/30 days

Headline L3.69 of 6 · top-3 weighted OWA over 13/16 pockets

How this is computed *Hide* ▾

Macro Stress

Last refreshed 5h ago

29 indicators elevated or worse

Live · Tier 1

AUTO-REFRESHED DAILY

SOFR

funding plumbing
5h ago

3.55% ↓

IORB-10bp

LEVEL 1 OF 6

CALM

via FRED

More ▾

EFFR

funding plumbing
5h ago

3.63% →

IORB-2bp

LEVEL 2 OF 6

WATCHING

via FRED

More ▾

3M T-BILL

treasury market
5h ago

3.60% →

FedFunds-3bp

LEVEL 1 OF 6

CALM

via FRED

More ▾

10Y YIELD

treasury market
5h ago

4.59% ↑

+21bp/5d

LEVEL 3 OF 6

ELEVATED

via FRED

More ▾

2Y YIELD

treasury market
5h ago

4.09% ↑

+19bp/5d

LEVEL 3 OF 6

ELEVATED

30Y YIELD

treasury market
5h ago

5.12% ↑

+17bp/5d

LEVEL 3 OF 6

ELEVATED

2S10S CURVE

treasury market
5h ago

50bp ↓

steepening

LEVEL 1 OF 6

CALM

HY OAS

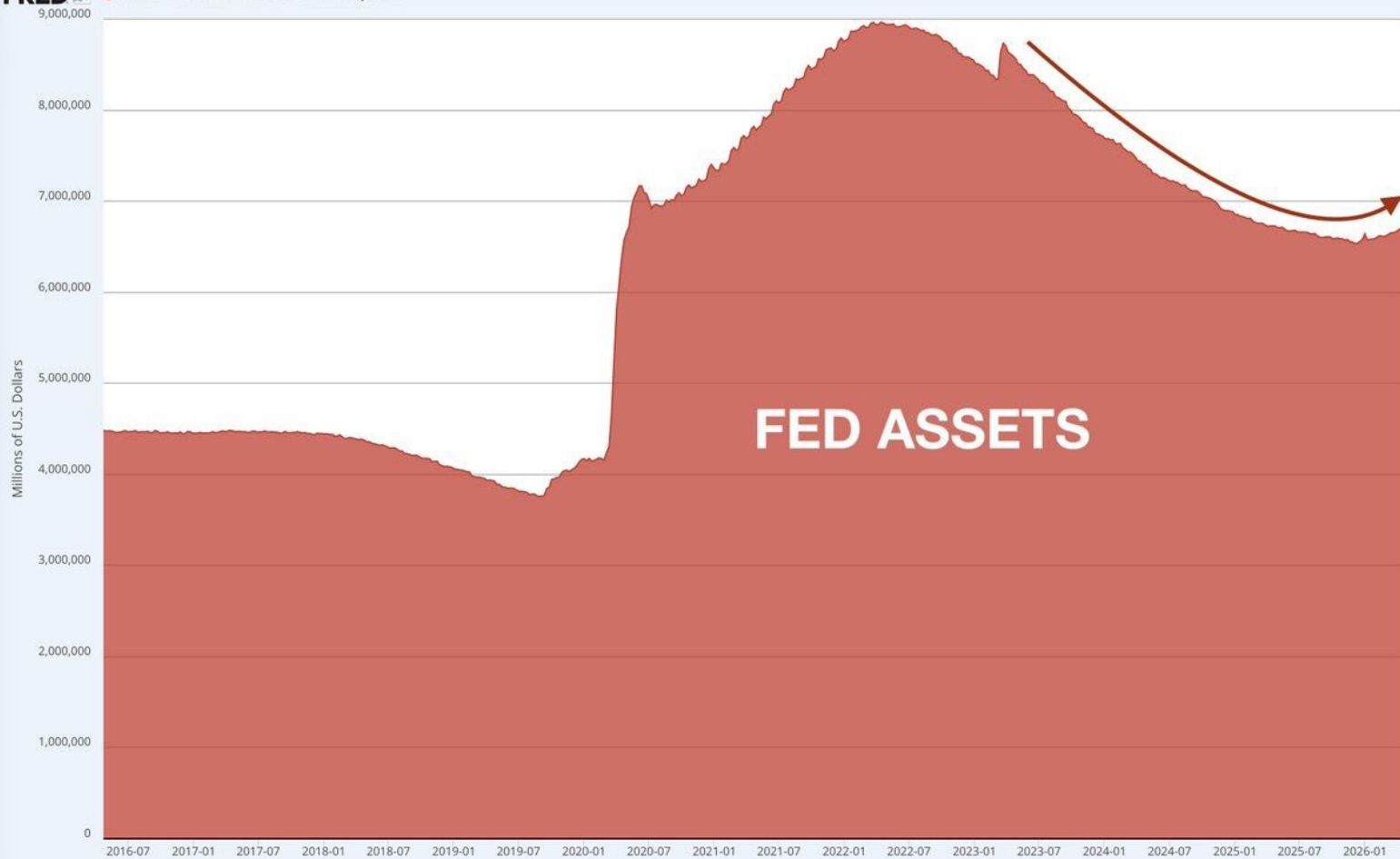
credit
5h ago

280bp ↓

-1bp/5d

LEVEL 1 OF 6

CALM





Simon Dixon ✓
@SimonDixonTwitt

X.com

Well, inflation can be offset by AI and robotics-driven productivity.

The side effect of that is unemployment, with the real economy experiencing deflation, while stocks concentrate wealth and create asset price inflation.

The real story is wealth transfer and the concentration of power.

That's why only asset holders survive.



Lanimal @Lanimal94843251 · 24m

Yep. Massive inflation incoming next few years.

Bitcoin Price Update

- **Key takeaway:**
 - Bottom is in, breakout isn't
 - DCA the chop.
 - The structural story (Iran/sovereign demand, MVRV cheap, power law intact) is stronger than it's been in 3 years
 - Short-term price action is noise
- \$60K February bottom locked in
 - ~50% confidence
 - 20-25% chance of mid-\$50K retest
- Just rejected off the 200-day MA at ~\$79K
 - Slope still declining = no "categorical bull" yet
- \$81-85K = the wall (major historical pivot)
- Mid-\$70Ks pullback expected in ~4 weeks
- Mid-June = pivot point, then rally resumes
- \$60K = structural floor (thesis breaks below)



Benjamin Cowen 
@benjamincowen

X.com

Bitcoin rejected off its 200D



22:17 · 5/18/26 · **361K** Views

Bitcoin History: Bitcoin Pizza Day

Bitcointalk Forum Post

laszlo

(OP)

Full Member



Activity: 199

Merit: 2490



Re: Pizza for bitcoins?

May 22, 2010, 07:17:26 PM

Merited by EFS (100), fillippone (62), paxmao (10), vizique (10), o_e_l_e_o (4), icopress (4), dragonvslinux (2), xtraelv (2), Farul (2), vapourminer (1), cAPSLOCK (1), bitbollo (1), ABCbits (1), Searing (1), DdmrDdmr (1), Husna QA (1), BitcoinFX (1), DireWolfM14 (1), 600watt (1), TheBeardedBaby (1), Toxic2040 (1), Heisenberg_Hunter (1), nullama (1), Spray. (1), TotSamiy (1), AricoIn (1), dektos (1)

#11

I just want to report that I successfully traded 10,000 bitcoins for pizza.

Pictures: <http://heliacal.net/~solar/bitcoin/pizza/>

Thanks jercos!

BC: 157fRqAKrDyGHR1Bx3yDxeMv8Rh45aUet

laszlo (OP)

Full Member



Activity: 199

Merit: 2490

Pizza for bitcoins?

May 18, 2010, 12:35:20 AM

Merited by EFS (100), DaRude (100), krogthmanhattan (100), fillippone (100), d5000 (55), nutildah (53), Ognasty (50), vapourminer (50), BlackHatCoiner (50), Seccour (50), debuni (50), Symmetrick (50), bitmover (40), hugeblack (28), Vod (20), sapta (20), icey (15), alani123 (12), LFC_Bitcoin (11), onurgozuek (11), Alex077 (10), TimtheYoutuber (10), Nomad88 (10), Totscha (10), the_poet (10), mnighwaffle (10), arthurbonora (10), leps (10), loreRex (10), suchmoon (9), ABCbits (9), LoyceV (8), tyz (7), cheefbuza (7), Betwrong (5), bitbollo (5), ebliever (5), mia_houston (5), LiteBit (5), klondike_bar (3), Farul (3), cygan (2), gbianchi (2), Halab (2), BitcoinFX (2), ChiBitCTy (2), MoparMiningLLC (2), bones261 (2), elianite (2), crypto_curious (2), bbigart (2), nikolaspaolo (2), dunnerjunge (2), ivaxmm (2), Cyrus (1), JayJuanGee (1), malevolent (1), batang_bitcoin (1), coolcoinz (1), Julien_Olynpic (1), sabotag3x (1), HI-TEC99 (1), AlcoHoDL (1), mole0815 (1), DdmrDdmr (1), Paolo.Demidov (1), Husna QA (1), TheQuin (1), DireWolfM14 (1), JanEmil (1), o_e_l_e_o (1), iluvbitcoins (1), UnDerDoG81 (1), Porfirii (1), GazetaBitcoin (1), S3cco (1), amishmanish (1), jamyr (1), BobLawblaw (1), Toxic2040 (1), dragonvslinux (1), digit (1), imstillthebest (1), Astargath (1), bonfire66 (1), lukax8 (1), gaston castano (1), jackthekeing (1), frankenmint (1), bitart (1), nullama (1), Crypto Library (1), naikturun (1), bubbalex (1), apoorvathey (1), BTCLiz (1), VB1001 (1), pushups44 (1), taserz (1), chimk (1), polymerbit (1), Financisto (1), nullius (1), SimpleFX (1), invincible49 (1), fishfishfish313 (1), tim-bc (1), thirdprize (1), Toughit (1), PascalCoin (1), M-BTC (1), texv (1), barjan (1), ajaxtempest (1), dektos (1), lonchafina (1), highalch (1), grinbuck (1), alia (1), inkling (1), Blocktables (1), Kda2018 (1)

#1



I'll pay 10,000 bitcoins for a couple of pizzas.. like maybe 2 large ones so I have some left over for the next day. I like having left over pizza to nibble on later. You can make the pizza yourself and bring it to my house or order it for me from a delivery place, but what I'm aiming for is getting food delivered in exchange for bitcoins where I don't have to order or prepare it myself, kind of like ordering a 'breakfast platter' at a hotel or something, they just bring you something to eat and you're happy!

I like things like onions, peppers, sausage, mushrooms, tomatoes, pepperoni, etc.. just standard stuff no weird fish topping or anything like that. I also like regular cheese pizzas which may be cheaper to prepare or otherwise acquire.

If you're interested please let me know and we can work out a deal.

Thanks,

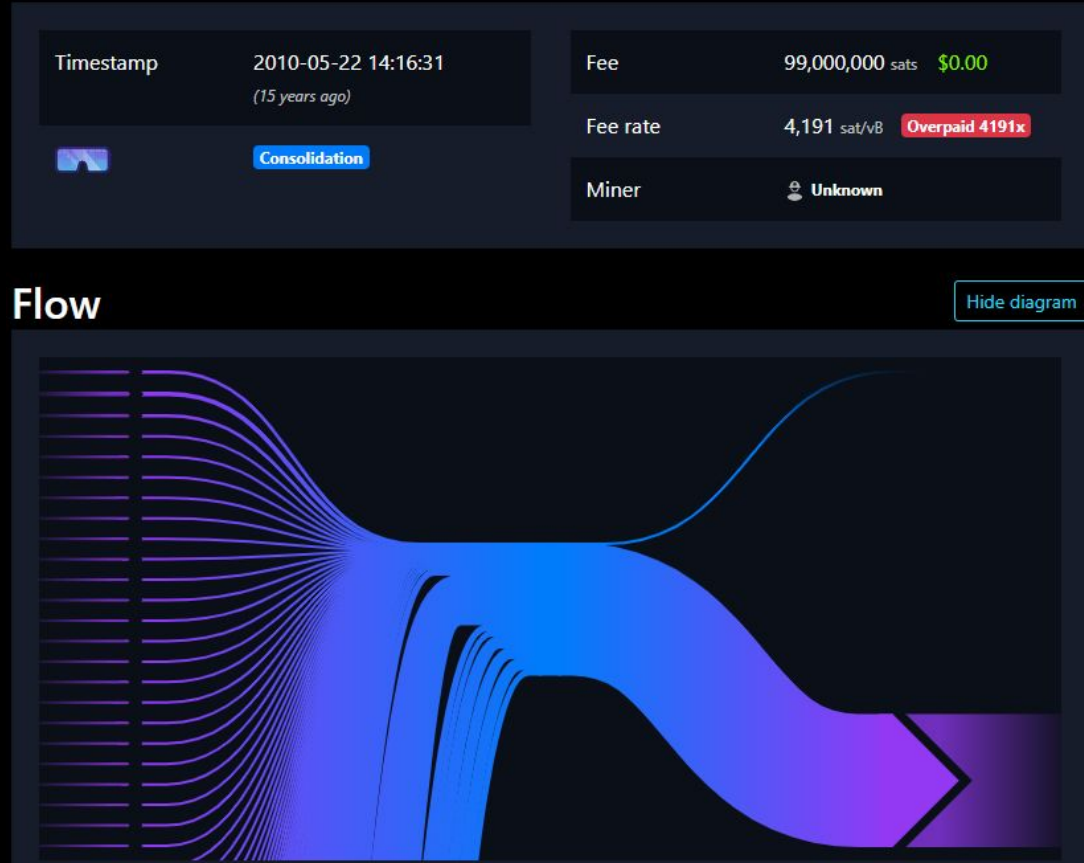
Laszlo

BC: 157fRqAKrDyGHR1Bx3yDxeMv8Rh45aUet

Bitcoin History: Bitcoin Pizza Day

The 10,000 BTC transaction was the only tx in block 57,043

It was the first known real world purchase made in bitcoin





JUST IN:  Iran launches Bitcoin-backed insurance service for shipping companies wanting to transit the Strait of Hormuz.

[@WatcherGuru](#)



FinancialFreedom  @FinFreedom414 · May 6



The inventor of Bollinger Bands just took a position in Bitcoin...and you're bearish.



John Bollinger  @bbands · May 6

Yesterday our trend model for Bitcoin turned positive and we took a position in our Tactica program, which is now fully invested. \$BTC





BREAKING: SpaceX reports they own 18,712 bitcoin worth more than \$1.45 billion — SEC filing 🙄

Note 7 - Digital Assets

Digital assets consist of the following:

(in millions except units of digital assets)	December 31,					
	2025			2024		
	Units	Cost Basis	Fair Value	Units	Cost Basis	Fair Value
Digital assets held:						
Bitcoin	18,712	\$ 661	\$ 1,637	18,712	\$ 661	\$ 1,749
Total	18,712	\$ 661	\$ 1,637	18,712	\$ 661	\$ 1,749



BITCOIN ADOPTION BY MAJOR U.S. BANKS

- ★ **U.S. Bank — October 2021**
(Bitcoin custody for institutional clients)
- ★ **PNC Bank — December 9, 2025**
(Direct spot Bitcoin trading for private banking clients)
- ★ **Bank of America — January 5, 2026**
(Advisors can recommend Bitcoin ETFs)
- ★ **Morgan Stanley — April 8, 2026**
(Launch of own spot Bitcoin ETF - MSBT)
- ★ **Goldman Sachs — April 14, 2026**
(Filed for Bitcoin Premium Income ETF)



Teddy Fusaro ✓

@teddyfuse



It always amazes me that the Satoshi statue is on the ground floor at the @NYSE



1:52 PM · May 4, 2026 · **11.8K** Views

Meme of the month



Bitmund Freud ✓
@BitmundFreud

X.com

When you're balls deep in Bitcoin... but nobody in real life has a clue.



From Jay Anderson ✓

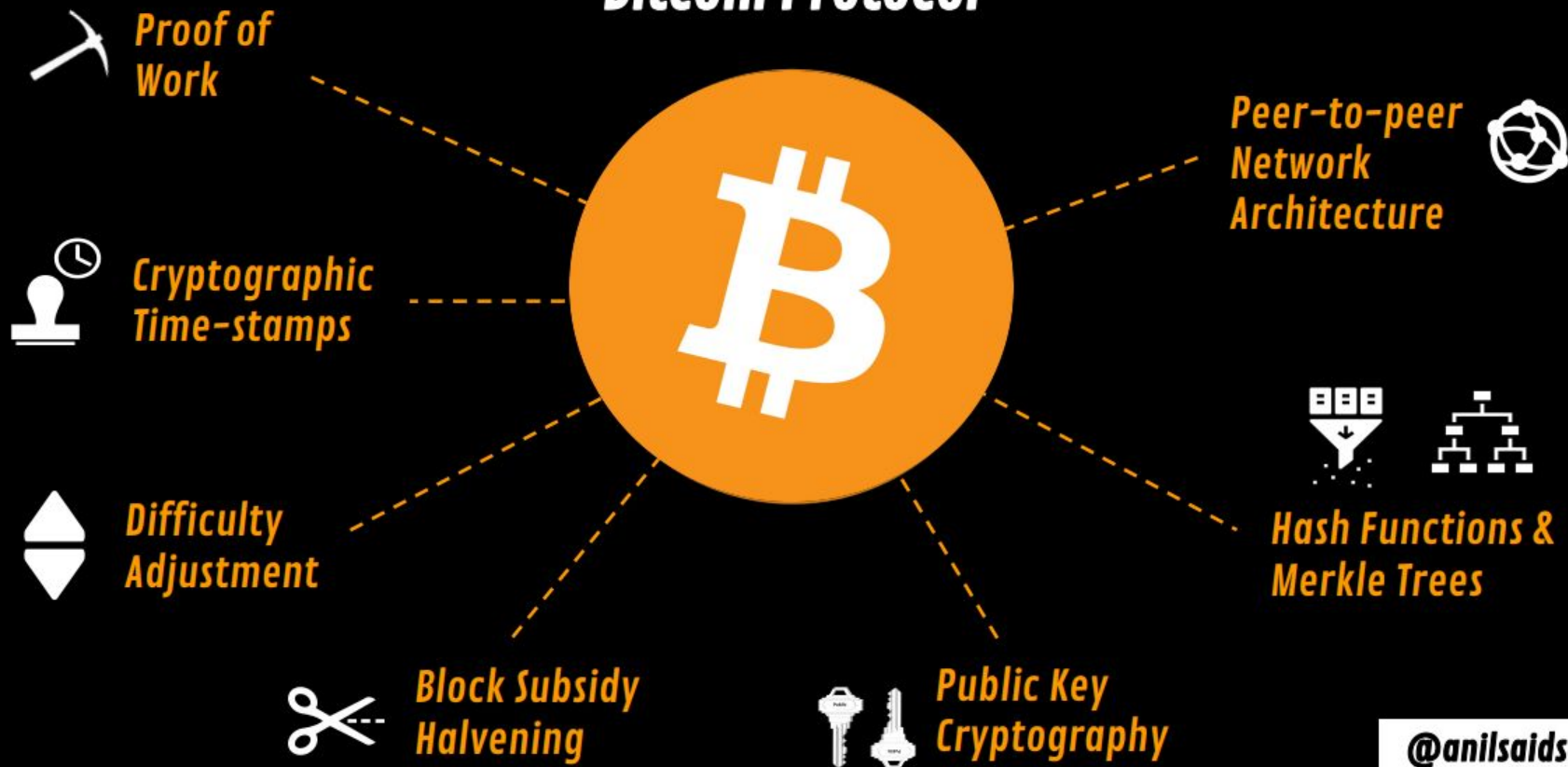
15:12 · 4/20/26 · 55K Views



How Does Bitcoin Work?

Why should you learn?

Mechanics of the Bitcoin Protocol



Bitcoin Network Concepts:

- Open source software
- The Bitcoin Protocol
- The Peer to Peer network

Transactions:

- UTXO's
- Public/Private Key Cryptography

Mining Prerequisites:

- The Mempool
- Blocks
- Hashing/SHA256
- Nonces
- Merkel Trees
- Block headers
- Linking blocks into a chain
- Cryptographic Time Stamping
- Hexadecimal format

Bitcoin Mining:

- Difficulty target
- Proof of work
- Mining Pools
- Chain verification
- Orphan chains and chain splits
- Block reward
- Nakamoto consensus
- 51% Attacks
- Nash Equilibrium
- How mining secures the network

Monetary Policy Concepts:

- Difficulty adjustment
- Halving and Block Subsidy

Is there any topic YOU would like to learn more about?

Mining Prerequisites:

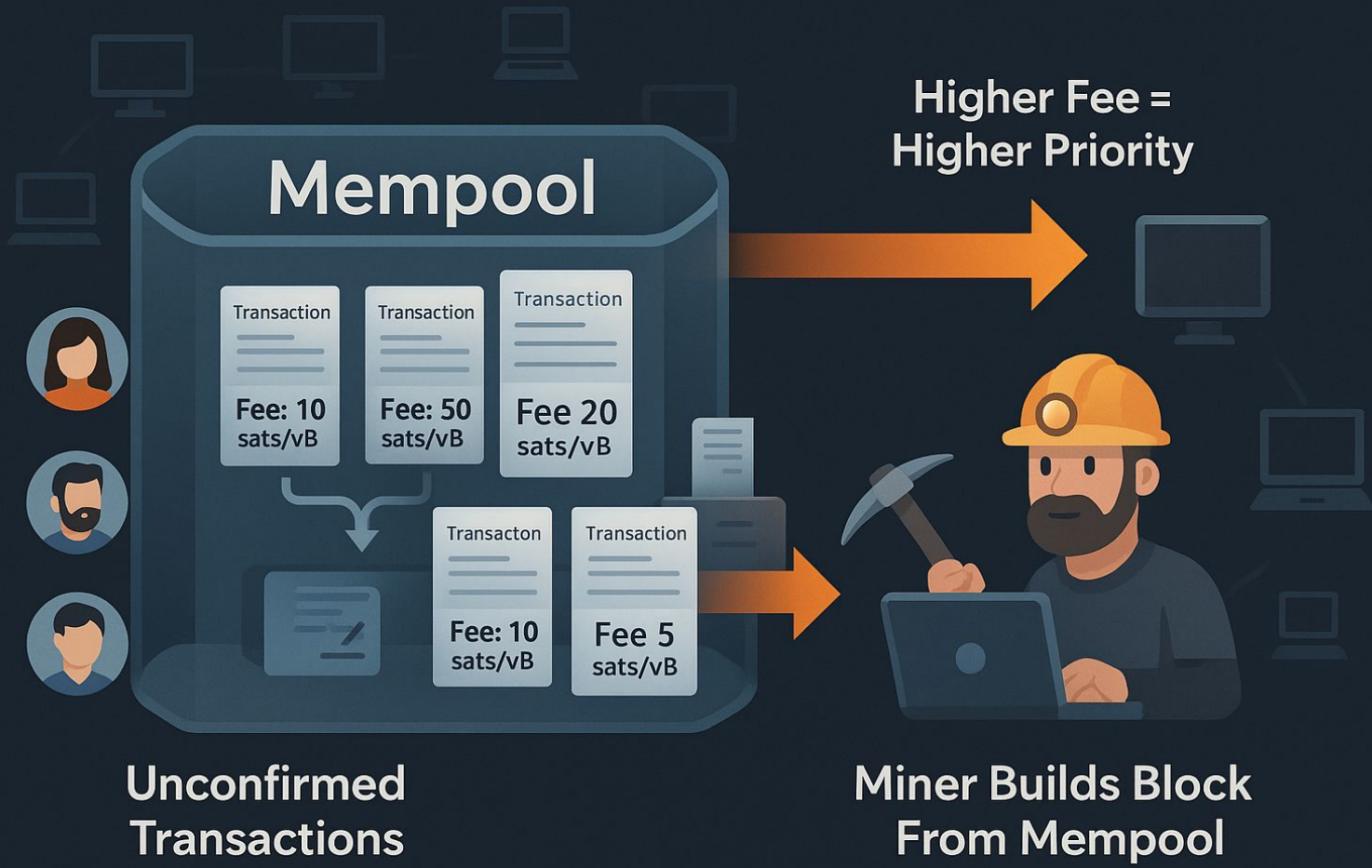
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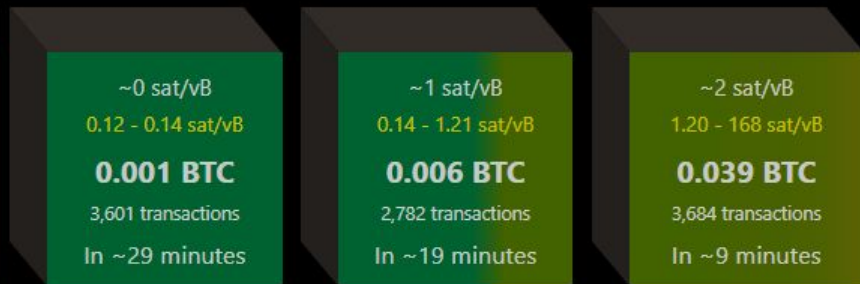
Bitcoin Mining:

- Difficulty target
- Proof of work



The Mempool

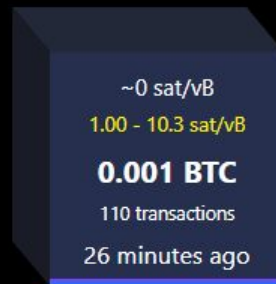




TRANSACTION FEES

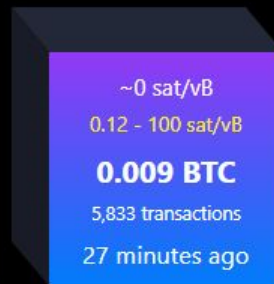


925908



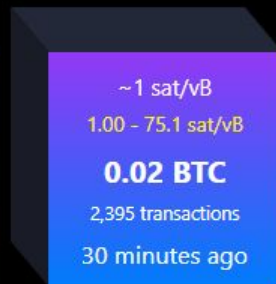
F2Pool

925907



AntPool

925906



SECPool



DIFFICULTY ADJUSTMENT

difficulty | halving

No Priority

Low Priority

Medium Priority

High Priority

1 sat/vB

\$0.13

1 sat/vB

\$0.13

1 sat/vB

\$0.13

3 sat/vB

\$0.38

~9.7 minutes

Average block time

▲ 3.63 %

Previous: ▼ 1.95 %

In ~9 days

December 10 at 9:05 AM

Blocks



— Tx 1
— Tx 2
— Tx 3

BLOCK

time-stamped
batch of confirmed
transactions
every 10 minutes on avg.

\$ 91,497.84

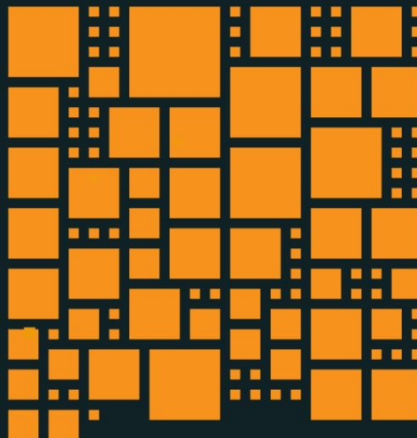
txid, block id or block height



Sponsored by

Satoshi
House

**Latest
Block:**
925,908
2025-11-30
14:42
Ø
39,437,910
≈ \$ 3.61M
47,172
bytes
110
transactions
avg fee rate
2.88
sats/vbyte

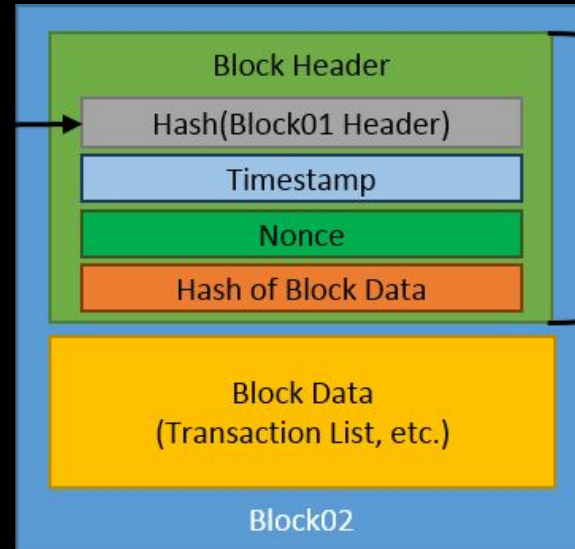
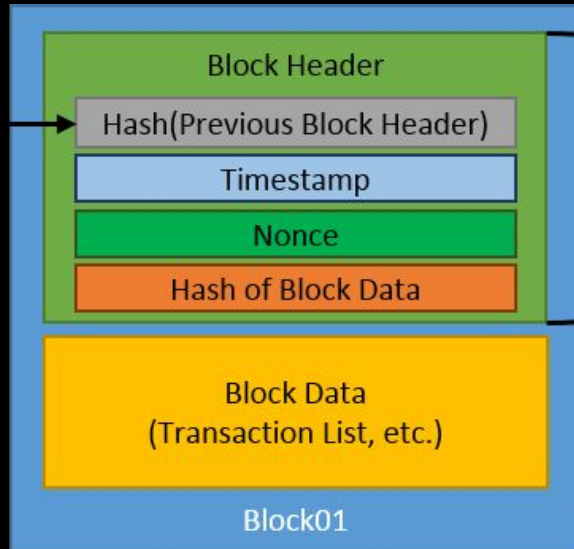


Mempool: 35,969 unconfirmed



The structure of a block

Size	Field	Description
4 bytes	Block Size	The size of the block, in bytes, following this field
80 bytes	Block Header	Several fields form the block header
1–3 bytes (compactSize)	Transaction Counter	How many transactions follow
Variable	Transactions	The transactions recorded in this block



Hexadecimal

Decimal System

0 1 2 3 4 5 6 7 8 9 → BASES

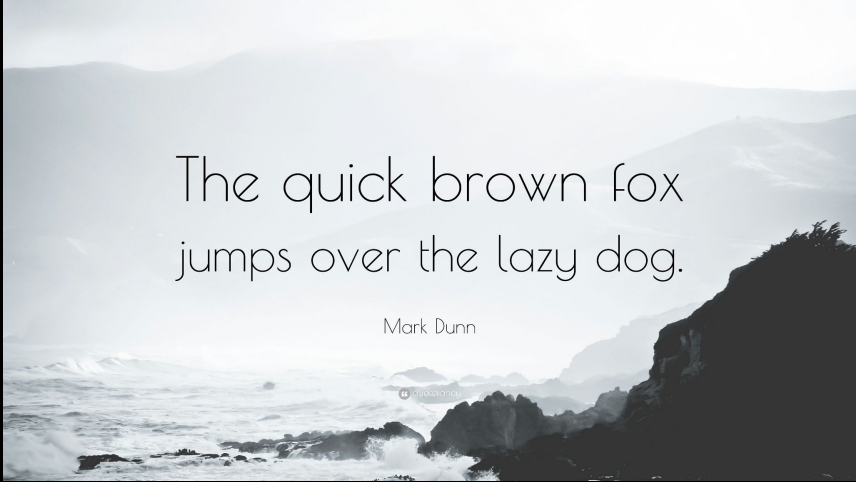
0 1 2 3 4 5 6 7 8 9 → Equivalent quantities

Hexadecimal System

0 1 2 3 4 5 6 7 8 9 A B C D E F

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Cryptographic Hash Functions



The quick brown fox
jumps over the lazy dog.

Mark Dunn

 quotealot



**The quick brown fox
jumps over the lazy dog.**

Mark Dunn

 quotefancy

Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

1. Introduction

Commerce on the Internet has come to rely almost exclusively on financial institutions serving as trusted third parties to process electronic payments. While the system works well enough for most transactions, it still suffers from the inherent weaknesses of the trust based model. Completely non-reversible transactions are not really possible, since financial institutions cannot avoid mediating disputes. The cost of mediation increases transaction costs, limiting the minimum practical transaction size and cutting off the possibility for small casual transactions, and there is a broader cost in the loss of ability to make non-reversible payments for non-reversible services. With the possibility of reversal, the need for trust spreads. Merchants must be wary of their customers, hassling them for more information than they would otherwise need. A certain percentage of fraud is accepted as unavoidable. These costs and payment uncertainties can be avoided in person by using physical currency, but no mechanism exists to make payments over a communications channel without a trusted party.

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123DtkHdTi2FHtLwwWoKJb9GgMmWpZW2va 0.00114000 BTC

ScriptSig (ASM) OP_PUSHBYTES_72 3045022100f1129e711a81f7fabdfbbcb4878c503130141825411e4f033105e04dceda692e702202b24787d65a621d6ddaf0b307f99aacee531c468786421a2939b5f9a32a33e8401
OP_PUSHBYTES_33 03a1c6e84948aa5961db717e9ec9b6990aeb80d6222384e6e8a42a2ef37949b461

ScriptSig (HEX) 483045022100f1129e711a81f7fabdfbbcb4878c503130141825411e4f033105e04dceda692e702202b24787d65a621d6ddaf0b307f99aacee531c468786421a2939b5f9a32a33e84012103a1c6e84948aa5961db717e9ec9b6990aeb80d6222384e6e8a42a2ef37949b461

nSequence 0xffffffff

Previous output script OP_DUP
OP_HASH160
OP_PUSHBYTES_20 0b63b834cb233aaace37200f1326fe37947a17e9
OP_EQUALVERIFY
OP_CHECKSIG

Previous output type P2PKH

bc1quncp8pcunasyv2qmvj5dw9l2k... wj8mlk04 0.00108790 BTC

ScriptPubKey (ASM) OP_0
OP_PUSHBYTES_20 e4f013871c9f6046281b64a8d717eab3a2b9f1d2

ScriptPubKey (HEX) 0014e4f013871c9f6046281b64a8d717eab3a2b9f1d2

Type V0_P2WPKH

123DtkHdTi2FHtLwwWoKJb9GgMmWpZW2va 0.00114000 BTC

ScriptSig (ASM) OP_PUSHBYTES_72 3045022100f1129e711a81f7fabdfbbcb4878c503130141825411e4f033105e04dceda692e702202b24787d65a621d6ddaf0b307f99aacee531c468786421a2939b5f9a32a33e8401
OP_PUSHBYTES_33 03a1c6e84948aa5961db717e9ec9b6990aeb80d6222384e6e8a42a2ef37949b461

ScriptSig (HEX) 483045022100f1129e711a81f7fabdfbbcb4878c503130141825411e4f033105e04dceda692e702202b24787d65a621d6ddaf0b307f99aacee531c468786421a2939b5f9a32a33e84012103a1c6e84948aa5961db717e9ec9b6990aeb80d6222384e6e8a42a2ef37949b461

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OP_CHECKSIG

Previous output type P2PKH

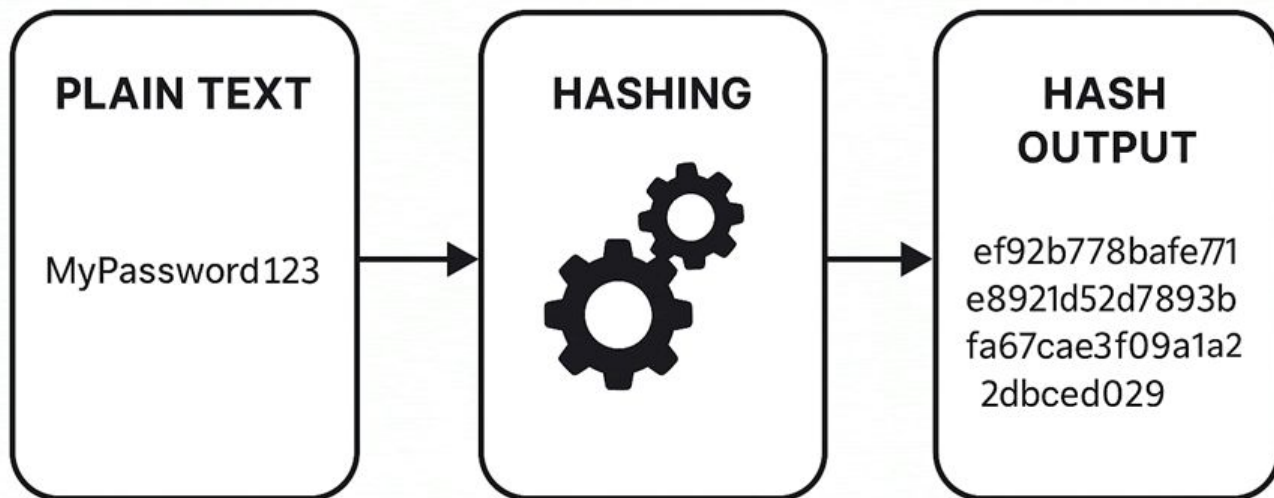
bc1quncp8pcunasyv2qmvj5dw9l2k... wj8mlk04 0.00108790 BTC

ScriptPubKey (ASM) OP_0
OP_PUSHBYTES_20 e4f013871c9f6046281b64a8d717eab3a2b9f1d2

ScriptPubKey (HEX) 0014e4f013871c9f6046281b64a8d717eab3a2b9f1d2

Type V0_P2WPKH

SHA-256





Satoshi Nakamoto



SHA

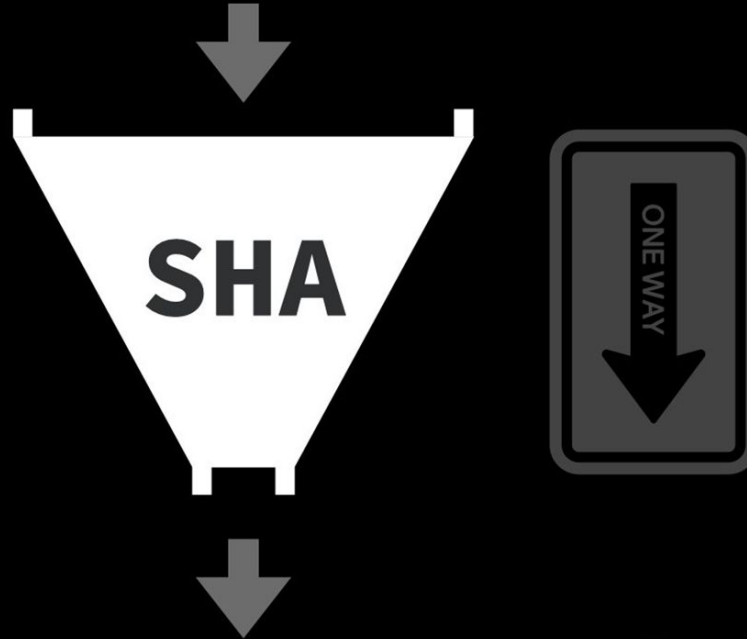


2662d47e3d692fe8c2cdb70b907ebb12
b216a9d9ca5110dd336d12e7bf86073b





Announcing the first release of Bitcoin, a new electronic cash system that uses a peer-to-peer network to prevent double-spending. It's completely decentralized with no server or central authority. See bitcoin.org for screenshots.



4aaef1e0a05028aabf60c929824a221f
9550d77e2a7824d9efd747a6335ff897



SHA256 Hash

Data:

Hash:

e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

Nonces

Transaction Configuration	Block: #3
	Timestamp: 1520610712
	Nonce: 4059
	Data:
	Kirill -> Hadelin 0.5 BTC
	Kirill -> Hadelin 0.7 BTC
	Hadelin -> Kirill 1.1 BTC
	Prev. Hash: 0000DF2E57FB432A
	Hash: 0000C4156AE3D5F7

```
Version
Previous Block ID
Merkle Root
Time
Bits
Nonce = 1
```

Block

Block:

#

1

Nonce:

72608

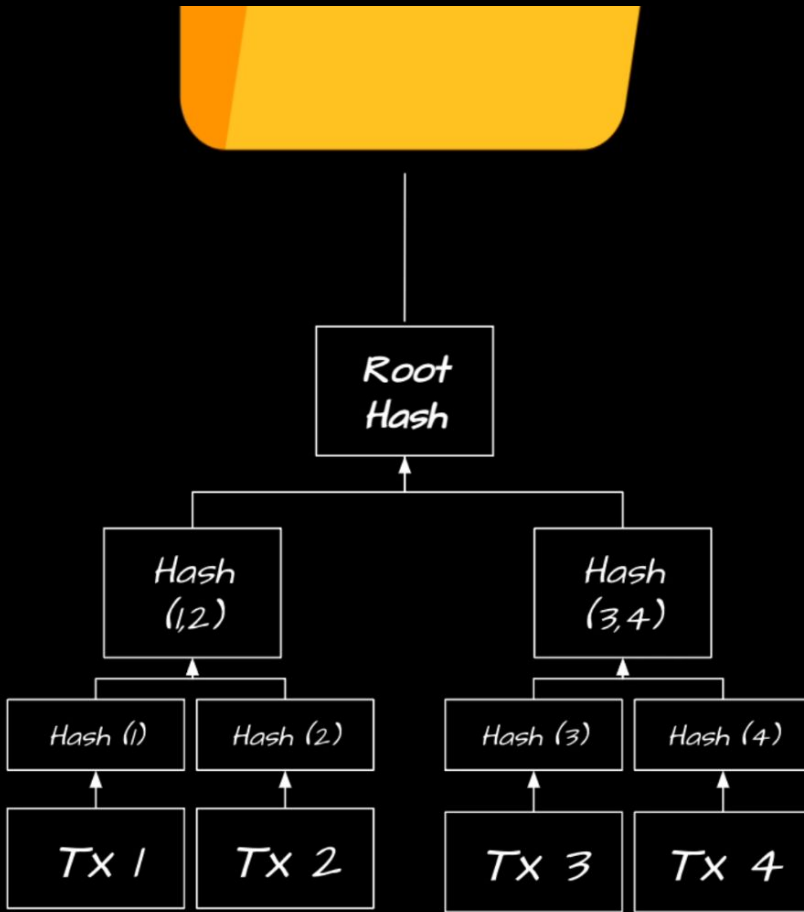
Data:

Hash:

0000f727854b50bb95c054b39c1fe5c92e5ebcfa4bcb5dc279f56aa96a365e5a

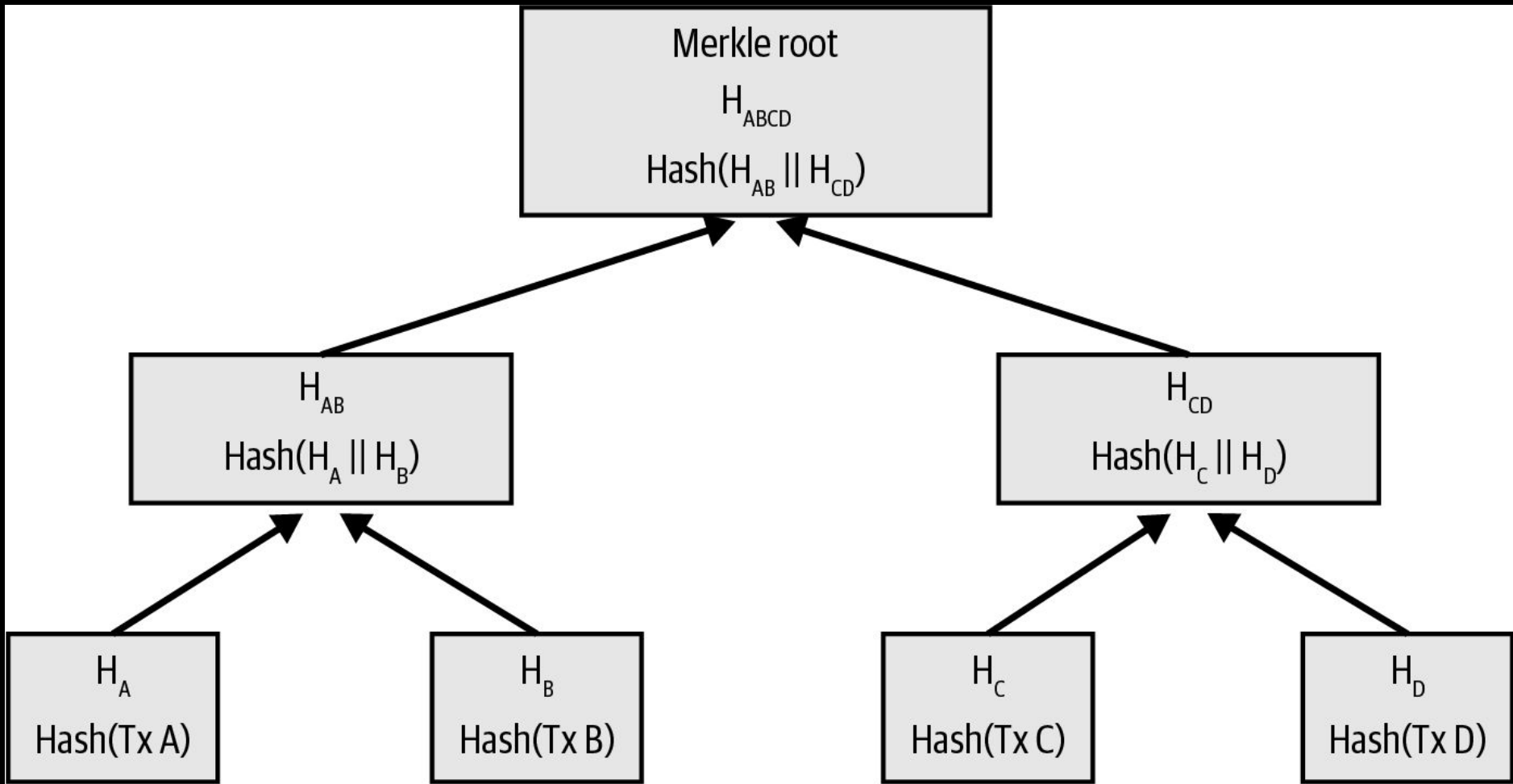
Mine

Merkle Trees



MERKLE TREE

data structure that
helps reduce
storage space and
easily prove
transaction validity



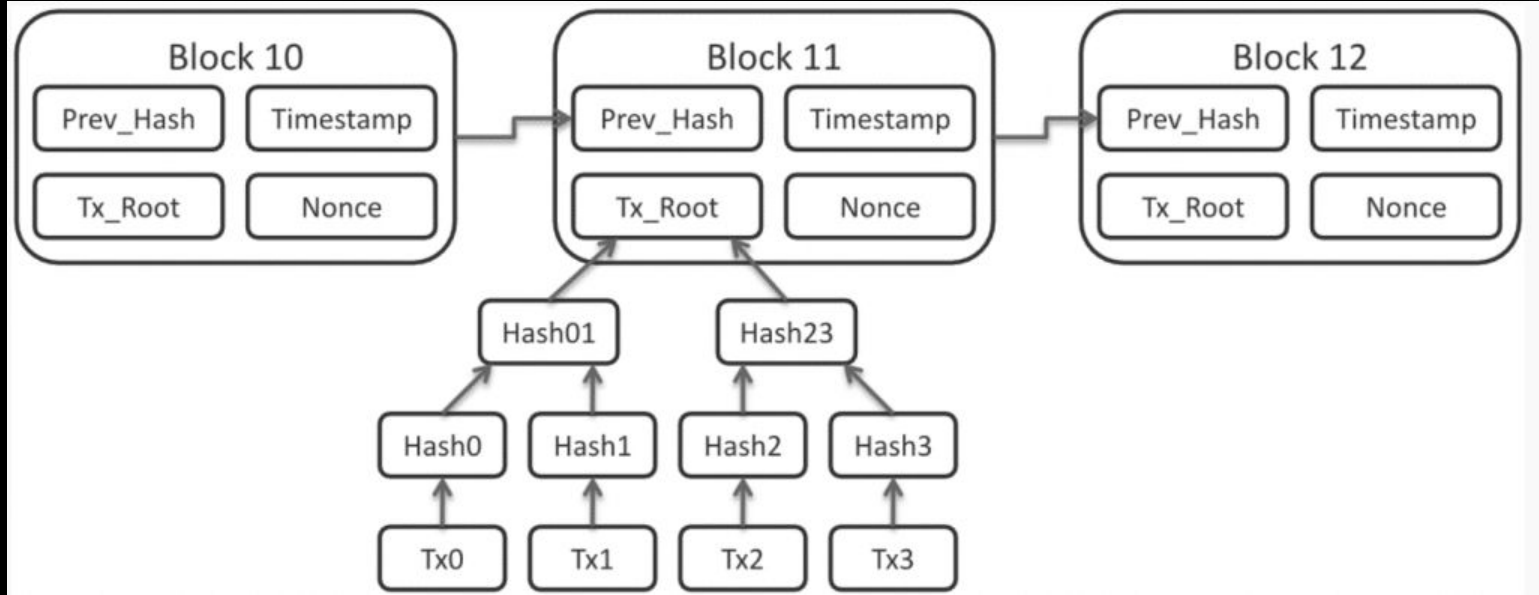
SHA256 Hash

Data:

Hash:

e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

Block Headers



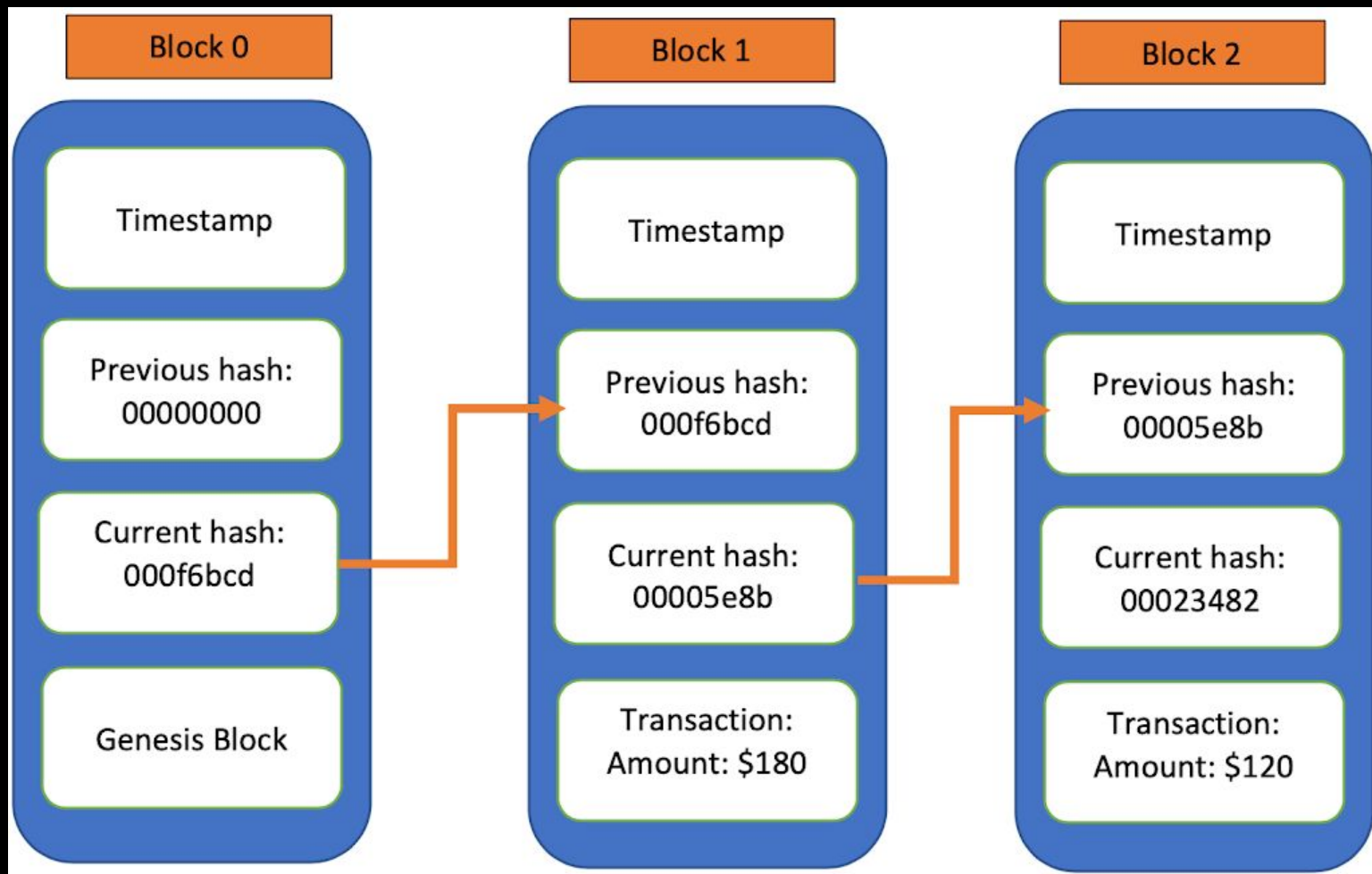
The structure of the block header

Size	Field	Description
4 bytes	Version	A multipurpose bitfield
32 bytes	Previous Block Hash	A reference to the hash of the previous (parent) block in the chain
32 bytes	Merkle Root	A hash that is the root of the merkle tree of this block's transactions
4 bytes	Timestamp	The approximate creation time of this block (seconds from Unix Epoch)
4 bytes	Target	The proof-of-work algorithm target for this block
4 bytes	Nonce	A counter used for the proof-of-work algorithm

```
{
  "size" : 43560,
  "version" : 2,
  "previousblockhash" :
    "00000000000000027e7ba6fe7bad39faf3b5a83daed765f05f7d1b71a16322
  "merkleroot" :
    "5e049f4030e0ab2debb92378f53c0a6e09548aea083f3ab25e1d94ea1155e2
  "time" : 1388185038,
  "difficulty" : 1180923195.25802612,
  "nonce" : 4215469401,
  "tx" : [
    "257e7497fb8bc68421eb2c7b699dbab234831600e7352f0d9e6522c7cf3f6c
    "[... many more transactions omitted ...]",
    "05cfd38f6ae6aa83674cc99e4d75a1458c165b7ab84725eda41d018a091766
  ]
}
```



Linking Blocks



The diagram illustrates the Bitcoin block structure and the process of creating a new block. It shows two blocks, each containing a previous block header hash, timestamp, difficulty, nonce, and merkle root. Transactions are hashed into the merkle root. A curved arrow indicates the process of creating a new block from the previous one.

Block Height 277316
Header Hash:
0000000000000001b6b9a13b095e96db
41c4a928b97ef2d944a9b31b2cc7bdc4

Previous Block Header Hash:
000000000000002a7bbd25a47c0374
cc55261021e8a9ca74442b01284f0569
Timestamp: 2013-12-27 23:11:54
Difficulty: 1180923195.26
Nonce: 924591752
Merkle Root: c91c008c26e50763e9f548bb8b2
fc323735f73577effbc55502c51eb4cc7f2e

Transactions

Block Height 277315
Header Hash:
0000000000000002a7bbd25a47c0374
cc55261021e8a9ca74442b01284f0569

Previous Block Header Hash:
00000000000000027e7ba6fe7bad39fa
f3b5a83daed765f05f7d1b71a1632249
Timestamp: 2013-12-27 22:57:18
Difficulty: 1180923195.26
Nonce: 4215469401
Merkle Root: 5e049f4030e0ab2debb92378f5
3c0a6e09548aea083f3ab25e1d94ea1155e29d

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Merkle Root: 5e049f4030e0ab2debb92378f5
3c0a6e09548aea083f3ab25e1d94ea1155e29d

Transactions

Blockchain

Block:

#

1

Nonce:

11316

Data:

Prev:

000000000000000000000000000000

Hash:

000015783b764259d382017d91a36d

Mine

Block:

#

2

Nonce:

35230

Data:

Prev:

000015783b764259d382017d91a36d

Hash:

000012fa9b916eb9078f8d98a7864e

Mine

Block:

#

3

Nonce:

12937

Data:

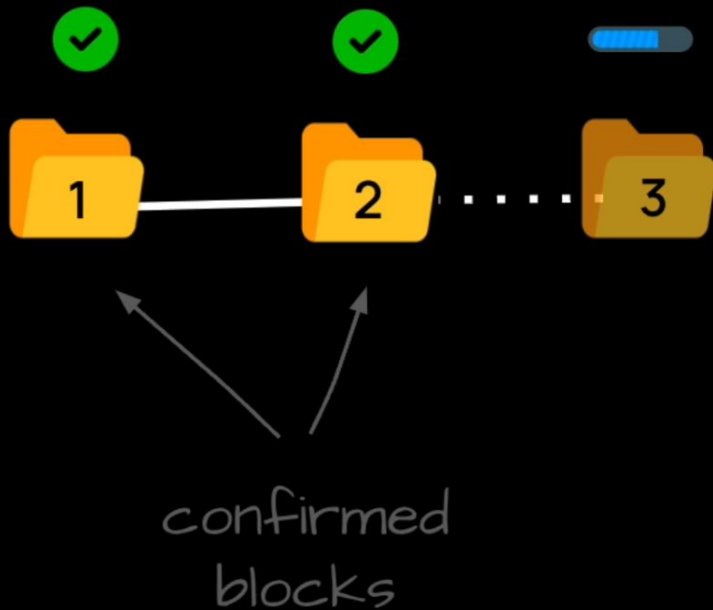
Prev:

000012fa9b916e

Hash:

0000b9015ce2a0

Mine



BITCOIN'S BLOCKCHAIN

sequentially-linked
blocks
historical record of all
confirmed transactions

Mining



Difficulty Target





00000000000000000000260572ca507ecfe8e4225c7449139333a4a29becf0a2e

0.96 ⑤ E→F

00000000000000000000de3eed8a0b33340034181200e6b285cf5d2f6019d25e

9.69 ⑤ B→E

0000000000000000000029d36ae6e7070890620f4ff04b93cc0982b7ad8cb0e8

1.33 ⑤ C→D

00000000000000000000c663b12f99daf6f0fca984149dffa1fb0b1427d497196

6.15 ⑤ B→C



SHA256 Hash

Data:

Hash:

e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855

Proof of Work

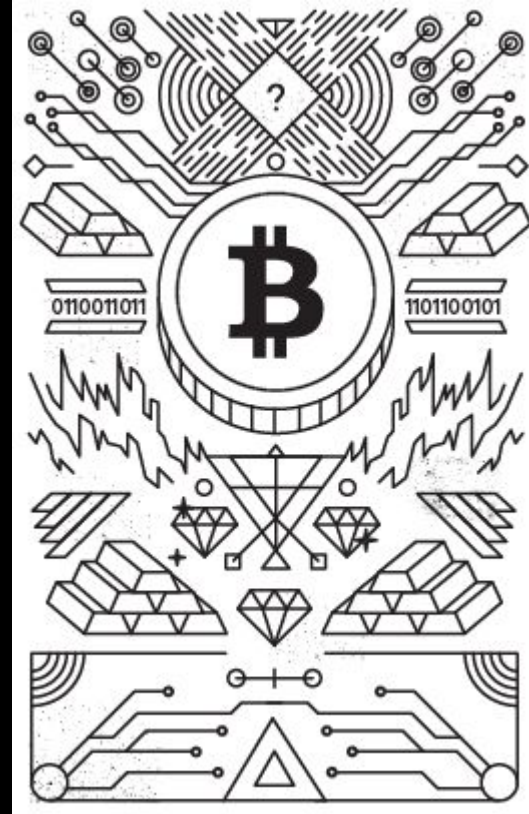
Any questions?

Mining Prerequisites:

- The Mempool
- Blocks
- Hashing/SHA256
- Nonces
- Merkel Trees
- Block headers
- Linking blocks into a chain
- Hexadecimal format

Bitcoin Mining:

- Difficulty target
- Proof of work



Bitcoin Network Concepts:

- Open source software
- The Bitcoin Protocol
- The Peer to Peer network

Transactions:

- UTXO's
- Public/Private Key Cryptography

Mining Prerequisites:

- The Mempool
- Blocks
- Hashing/SHA256
- Nonces
- Merkel Trees
- Block headers
- Linking blocks into a chain
- Cryptographic Time Stamping
- Hexadecimal format

Bitcoin Mining:

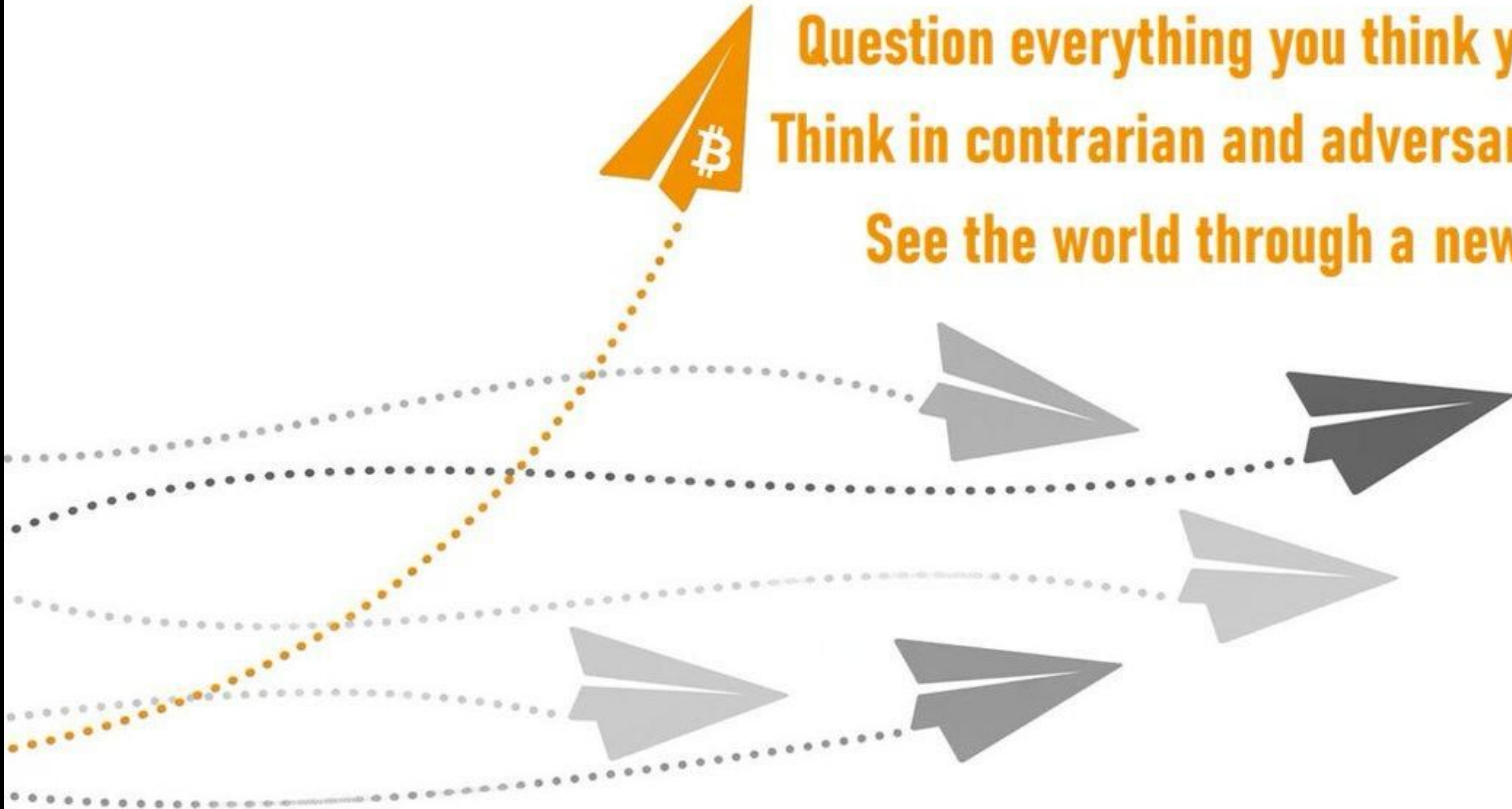
- Difficulty target
- Proof of work
- Mining Pools
- Chain verification
- Orphan chains and chain splits
- Block reward
- Nakamoto consensus
- 51% Attacks
- Nash Equilibrium
- How mining secures the network

Monetary Policy Concepts:

- Difficulty adjustment
- Halving and Block Subsidy

Bitcoiner Values

Question everything you think you know
Think in contrarian and adversarial terms
See the world through a new lens



Jersey City Bitcoin Education Center Library & Book Sharing Program

- Bring bitcoin books you have already read and exchange for ones you haven't! Trade back the following month.
- Borrow books from current library with Sats deposit. Return next month and receive your deposit back.
- Konsensus Network provides discounts on book purchases from Bitcoin Communities. This should be available soon.

Feedback!

- How do you like this format?
- Is any of the material too technical?
- What topics would you like discussed?

Appendix

Website!

- More inviting, more beginner
- Open to all
- Open house type of feel
- Help members turn to a bitcoin standard, step by step, guided way to hold their hands, it's coming up a lot on the show
 - The challenge is everyone is different
-

Freedom Tech Spotlight: Fedi

- What is Fedi?
- How is it different from Wallet of Satoshi or an on chain wallet?
- Goal of circular economy

Download Fedi

Apple



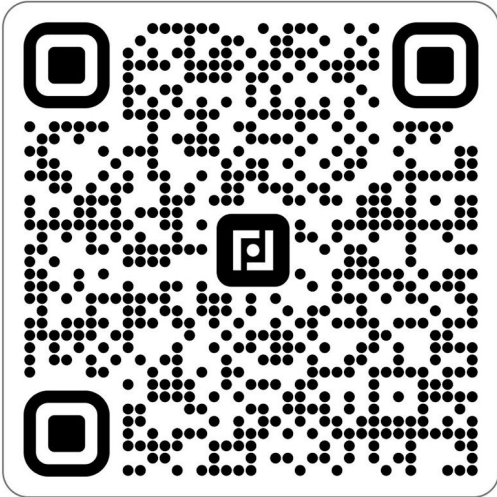
Android



Add Me!

Settings

×

A square QR code with a black and white pixelated pattern. In the center of the QR code is a small square icon containing a white 'P' on a black background. The QR code is enclosed in a thin black border.

<https://app.fedi.xyz/link?screen=user...>  Share

witty hedgehog #b6b5



JC BTC Group Chat



fedir:room:!!UCgMvqtgVjtcCwwSF:m1....  Copy

Community Invite





Thank You!